Venkata Lakshmi Parimala Pasupuleti

Agentic AI Engineer · LangGraph · RAG Pipelines · LLM Evaluation · Production AI · Financial Compliance · Python · AWS

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Agentic AI engineer with 8 years shipping production backend systems and AI pipelines in regulated environments. Built **Leasel**: an 8-agent LangGraph agentic AI platform (Triage → Retrieval → Reasoning → HITL Escalation) with production-grade RAG pipelines over regulated residential compliance data — 93.6% precision across 1,240 lease PDFs with deterministic auditability gates and human-in-the-loop review on all regulated decisions. Built **AgentProbe** (open-source): LLM evaluation and monitoring toolkit with a CI/CD regression harness, 5-metric eval engine, golden datasets, and multi-provider agent orchestration — used in production to prevent AI regressions on every commit. At Entrata: **7 years owning regulated financial data operations** — HUD income eligibility, HAP payment processing, rent compliance, and PHI-compliant document pipelines for 100K+ housing units (180K records, zero compliance violations, 99.5% uptime). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages & Frameworks: Python (expert), TypeScript, JavaScript, Java, Kotlin, SQL (expert — 200+ production optimizations, execution plan tuning, indexing, partitioning), PL/SQL, PHP, R, Bash; FastAPI, Spring Boot, Django, Node.js, React.js; gRPC (familiar), REST APIs, GraphQL, OpenAPI

Agentic AI & LLM: LangGraph, LangChain, Agentic AI Systems, Agent Orchestration, Multi-Agent Orchestration, LLM Orchestration, Tool Calling, HITL (Human-in-the-Loop), RAG Pipelines, Hybrid Search, Semantic Chunking, Citation Grounding, Auditability, LLM Evaluation, LLM-as-Judge, Eval-Driven Development, Regression Harnesses, Golden Datasets, LLMOps, Agent Observability, Hallucination Detection, Responsible AI, Guardrails, Prompt Injection Defense, Financial Services Compliance, ChromaDB, Qdrant, Vector Databases, OpenAI API, Anthropic API, Groq, Ollama, AWS Bedrock, Vertex AI

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue, Lambda, SageMaker, Step Functions, API Gateway, IAM, SQS, OpenSearch), GCP (Cloud Storage, Vertex AI — familiar), Kubernetes (K8s), Docker, Apache Kafka, RabbitMQ, Redis, CI/CD (GitHub Actions), Terraform, Microservices, Distributed Systems, Fault-Tolerant Systems, PostgreSQL, Snowflake, Apache Airflow, Celery, Prometheus, Grafana, OpenTelemetry

ML & Data: PyTorch, scikit-learn, XGBoost, Isolation Forest, Anomaly Detection, Feature Engineering, Statistical Modeling, NumPy, Pandas, Tableau, Power BI, MLflow, HuggingFace, BERT, spaCy, ETL/ELT Pipelines, Data Governance, Data Lineage

PROJECTS

AgentProbe: Open-Source LLM Evaluation & Agent Monitoring Toolkit

OPEN SOURCE · EVAL TOOLING

Python · LangGraph · FastAPI · ChromaDB · Docker · GitHub Actions · React · OpenAI · Anthropic · Groq · Ollama — github.com/venkatalakshmiparimala/AgentProbe

- Built a fully local, zero-cost **LLM evaluation and monitoring toolkit** with a **15-endpoint FastAPI backend**, **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated **regression harness with golden datasets**, multi-provider LLM routing (OpenAI/Anthropic/Groq/Ollama), and offline + online evaluation loops — deployed as open-source with **zero vendor dependency**.
- 3-agent LangGraph runtime (Orchestrator → RAG Tracer → Auto Fix) with tool calling and inter-agent state handoffs; React **agent observability dashboard** tracks per-run metrics, retrieval drift, and latency in real time; used in production to evaluate LeaseI across 1,240+ compliance test cases with CI/CD regression gates on every commit.

Leasel: Agentic AI Investigation Platform — Regulated Compliance, HITL, RAG

REGULATED DOMAIN · AGENTIC AI

Python · LangGraph (8-agent) · FastAPI · ChromaDB · TF-IDF · Hybrid Search · OCR · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an **8-agent LangGraph agentic AI system** with durable state for enterprise compliance investigation: Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → **Human Escalation** (HITL). Design rule enforced in code: LLM extracts and reasons; human or deterministic logic makes the final compliance call — every answer **cited to exact source clause or explicitly flagged as unsupported** (auditability by architecture).
- **Production-grade RAG pipeline**: OCR-fallback ingestion, section-aware chunking, **hybrid search** (TF-IDF + semantic reranking), citation grounding, missing/conflicting clause detection, conditional HITL escalation when confidence threshold not met. Achieved **93.6% precision, 87.9% F1** across 10 compliance rules on 1,240 residential lease PDFs (5 US states + federal); pass@1 81.7%, pass@3 88.3% via AgentProbe regression harness.

AI on FHIR: NLP-Driven Medical Records Query System

REGULATED AI · HEALTHCARE COMPLIANCE

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings — github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced manual clinical retrieval time ~60% by building a Flask + React NLP platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; enforced **HIPAA-compliant RBAC** across structured medical records — demonstrates regulated AI compliance pattern (HIPAA ≈ financial services data governance).

WORK EXPERIENCE

Blue Horse Digital

Sep 2026 – Present

AI/ML Engineer (Volunteer) — ML Models · Data Pipelines · AI Integration

- Developing ML models and data analysis pipelines for software applications and BI projects; integrating AI components with back-end and front-end teams; building pipelines that surface insights from raw business data into application-facing APIs.

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern — Agentic AI · RAG · LangGraph · Compliance Guardrails

- Built an **agentic mission-planning system** as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing); implemented RAG layer over FAA Part 107 corpus with cited sources; engineered **validation guardrails** enforcing regulatory compliance limits before export — non-compliant missions blocked with grounded rejection reasons.

George Mason University

Aug 2024 – Aug 2025

Research Associate — ML Modeling · Anomaly Detection · Analytics Engineering

- Built multi-source prediction pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest for automated data integrity checks across 3 research pipelines; designed Tableau/ggplot2 dashboards driving **40% improvement** in research decision comprehension; fine-tuned custom classification model (HuggingFace, PyTorch) with inference API on Quart.

Entrata — PropTech / Real Estate SaaS Sep 2018 – Aug 2024

Senior Software Engineer — Regulated Financial Operations · Compliance at Scale · AWS · Auditability

- **Owned the HUD income-limit compliance pipeline for 7 years** — federal source rows expanded into ~180,000 domain records powering income eligibility, rent calculations, HAP payment processing, and tax-credit compliance for 100K+ housing units. Every release included baseline validation, reconciliation, and rollback preparation because one wrong import means thousands of residents receive incorrect eligibility decisions. **Zero compliance violations across the entire ownership period.**
- Built and owned the **RabbitMQ-based async financial processing system** for gross-rent updates across thousands of tenant records — designed idempotency guarantees and retry logic eliminating double-processing failures; owned system through production incidents. **Optimized 200+ SQL queries (30% performance gain)** via execution plan analysis, indexing, partitioning; complex schemas for eligibility, financial, and compliance workflows across multi-tenant production environments (**99.5% uptime, 120+ Agile sprints**).
- Achieved **100% PHI compliance** on AWS S3 document migration with Libsodium-encrypted pipelines (zero security incidents); processed **1.6M HUD/Tax Credit records** via production AI pipeline (Python, PyTorch, OpenAI API) replacing manual workflows; led microservices Docker redesign (**50% scalability improvement, 30% deployment reduction**); eliminated 80% manual operational workload via Python/Bash automation; **300+ code reviews, 5+ engineers mentored**.

Entrata India Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Full-Stack · Data Pipelines · AWS S3 · Compliance Workflows

- Built full-stack property management modules (PHP, PostgreSQL, JavaScript); designed SQL import pipelines for 1.6M HUD/Tax Credit income limit records; developed compliance workflows and AWS S3 document integrations; participated in sprint planning, code reviews, and performance debugging.

EDUCATION

M.S. Data Analytics Engineering — George Mason University May 2026 · GPA 3.87

AI pipelines, ML systems, statistical modeling, data engineering, anomaly detection, data governance

B.Tech Computer Science & Engineering — JNTU Anantapur 2013 – 2017